Highlights

From Radiation Shielding to Spin Glasses: Auditing the Adjoint-Importance Toolkit for Rare-Event Sampling

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- Transport adjoint importance and the rare-event committor are identical; verified to machine precision.
- Weighted Ensemble validated against exact enumeration; two decades deeper than naive Monte Carlo.
- FW-CADIS attains equivalent threshold coverage at roughly one-eighth the allocation cost.
- Learned importance coordinate gives a 1.19× FOM ratio, bounded by a two-level bootstrap.
- Deep-ensemble follow-up resolves that ambiguity: a decisive 0.087× loss (95% CI $[0.051, 0.155]$), the paper's only interval excluding parity.
- At a depth where naive is provably blind, self-consistency training gives a significant reliability win: 66.7% vs. 40.0% ($p = 0.017$), the paper's one significant positive result.
- Uniform exact/validated/open grading maps where the transport toolkit transfers to spin systems.

From Radiation Shielding to Spin Glasses: Auditing the Adjoint-Importance Toolkit for Rare-Event Sampling

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ABSTRACT

Radiation transport solved rare-event sampling decades ago with the adjoint importance function and the CADIS/weight-window machinery, a quantity that turns out to be mathematically identical to the rare-event committor of Markov chain theory. We transfer this toolkit to statistical mechanics and test it end to end on the 2D Ising ferromagnet and the Edwards–Anderson spin glass, asking which parts of it actually earn their keep. The adjoint-committor identity is exact, verified here to machine precision, and Weighted Ensemble sampling (weight windows applied to spin configurations) reaches two decades deeper into the tail than naive Monte Carlo at matched cost. A learned importance coordinate trained on a milestoning surrogate label does not beat a hand-picked energy coordinate. An ensemble of independently-trained networks, built to resolve an earlier ambiguous result, instead converges on a statistically decisive loss (FOM ratio 0.087×, 95% CI $[0.051, 0.155]$). Every comparison up to that point, though, was run at a depth where naive Monte Carlo still holds its own. Tested at a depth calibrated so naive is provably blind, a self-consistency training objective shaped directly on the harmonic committor condition, rather than an energy-adjacent proxy of it, finds the rare event significantly more often than the hand-picked coordinate (66.7% vs. 40.0% hit rate across three independent networks, $p = 0.017$), though a matching cost-efficiency advantage has not yet been established. A global variance-reduction objective (FW-CADIS) matches uniform allocation's error profile at one-eighth the cost, but does not flatten relative error the way it is meant to. Naive Monte Carlo is not beaten outright at any lattice size tested ($L = 8$–32) at shallow threshold depths. Together, these results say the transport toolkit's value is concentrated in the genuinely rare regime it was built for, and depends on a training objective shaped like the true committor rather than a convenient proxy of it.

1. Introduction

The application domains on the statistical-mechanics side are not hypothetical: Weighted Ensemble, Forward Flux Sampling, and committer-based methods are already the working tools for nucleation and crystallization, protein folding and conformational transitions, and chemical reaction rates via transition-state theory [1–3]. A transport toolkit that reliably picks the importance function for them, rather than requiring a hand-chosen progress coordinate, would apply directly to that existing user base. The obstacle is structural, since an unbiased estimator of an event of probability p has relative error of order $(Np)^{-1/2}$, so each decade of rarity costs an order of magnitude in computing [1, 4]. Every serious remedy therefore replaces more samples with samples drawn, an idea present in Monte Carlo since importance sampling and trajectory splitting [5, 6].

Radiation transport met the problem first, in deep-penetration shielding, where analog scoring probabilities fall below 10^{-10} [7, 8]. Its remedy is the *adjoint* flux, or importance function, which runs the transport equation backwards from the detector response and measures the expected contribution of a particle at each phase-space point [9, 10]. An importance function so obtained is used both to bias source emission and to set weight windows, inside which particles are split and outside which they are killed by Russian roulette [5, 11]. CADIS automated this by deriving source biasing and weight windows consistently from a coarse deterministic adjoint, giving figure-of-merit gains of several orders of magnitude on realistic geometries [8, 12]; its forward-weighted extension FW-CADIS replaces the single-detector objective with a *global* one, seeking uniform uncertainty across a whole family of responses [13, 14]. Throughout, performance is judged by the figure of merit $\mathrm{FOM} = 1/(\sigma^2_{\mathrm{rel}} T)$, a cost-normalised inverse variance [7, 13]. Transition Path Theory describes the statistics of reactive trajectories through the probability of reaching a target set

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before returning to the origin set (the committor) $h(s)$, which is the optimal reaction coordinate in a precise sense [3, 15, 16]. Its practical samplers are splitting methods in disguise: Weighted Ensemble splits and merges weighted replicas across bins in a progress coordinate [17], a scheme later shown statistically exact for broad classes of processes and binnings [18]; Forward Flux Sampling and Adaptive Multilevel Splitting advance an ensemble through fixed or adaptive interfaces [1, 19]; and Transition Path Sampling works directly in trajectory space [20]. What unites and limits them all is dependence on a user-supplied coordinate, with no general prescription for choosing it [1, 2]. Lattice spins are the natural testbed: the 2D Ising ferromagnet [21] has an exact solution to validate against [22], while the Edwards–Anderson model adds quenched disorder, frustration and no obvious coordinate [23, 24]. Both are usually simulated with Metropolis dynamics [25], whose critical slowing down motivated cluster updates [26, 27] and flat-histogram schemes [28, 29], methods that flatten a free-energy landscape rather than steer trajectories toward a specified tally, and so optimise a mismatched objective here [29].

The identity between the two is classical rather than new. Both objects are instances of Doob's h-transform, which conditions a process on a rare terminal event by tilting its generator with the corresponding harmonic function [30]; the same construction reappears as the driven process of large-deviation theory and underpins the zero-variance sampler [4, 31], and again in stochastic-control form as the solution of a Gibbs variational principle [32]. Machine learning has since supplied a third route: physics-informed networks solve high-dimensional PDEs without a grid [33], and this was applied to the committor equation itself [34, 35], with active importance sampling added to concentrate training effort where the loss is dominated by rarely visited regions [36]. A separate, active line trains a committor variationally and uses it as a metadynamics-like collective variable with a transition-state-oriented bias potential, rather than a weight-window schedule, to sample molecular rare events uniformly across competing pathways [37], methodologically distinct from the splitting/rouletting machinery this paper transfers, though the same underlying object. Generative models were meanwhile trained directly on spin systems: variational autoregressive networks [38] and Boltzmann generators [39], with later work recovering unbiased observables by reweighting [40] or embedding the model in a Markov chain [41]. Closest to this work, a recent preprint learns an importance function on a discrete chain from a label-free self-consistency objective and explicitly identifies adjoint flux with committor [42].

The gap this leaves is specific. The identity is established [30, 31] and neural committors exist [34, 35, 37, 42], but the transport *toolkit* built around that identity (consistent weight windows, the global FW-CADIS objective, the figure of merit as an evaluation standard) was designed for precisely the regime where learned-committor methods are weakest [8, 13], and has never been transferred to spin systems and audited: every neural-committor line above samples by continuously reweighting or biasing dynamics, not by the transport toolkit's discrete population control (splitting, roulette, resampling to a fixed per-bin occupancy), so that mechanism and its FOM/FW-CADIS evaluation standard remain untested here. That is the project reported here, summarised in three claims:

1. The optimal importance function for a rare tally *is* the committor, and tilting by the exact committor is zero-variance [30, 31].

2. CADIS/FW-CADIS weight windows can tame the fragility of an imperfect importance estimate and, in the global form, flatten relative error across a family of rare thresholds [8, 13].

3. The FOM is a principled evaluation metric and a natural training objective, aligned with variance reduction rather than with free-energy surrogates [7, 29].

We test each of these three claims end to end on the 2D Ising ferromagnet and the Edwards–Anderson spin glass, treating each piece of machinery as a hypothesis rather than a technique to be demonstrated. Claim (i) holds to numerical precision; claims (ii) and (iii) proved more interesting, since several apparent wins did *not* survive a proper bootstrap confidence interval, and the paper's one result that did survive such scrutiny only became significant once the target was pushed rare enough that naive Monte Carlo could no longer compete at all. §3 reports every comparison in full.

2. Theoretical Approach

2.1. Theory: the adjoint=committor identity

2.1.1. Linear transport and the adjoint importance function

Let $P = (\mathbf{r}, \boldsymbol{\Omega}, E)$ be a phase-space point and $\psi(P)$ the particle flux obeying the linear transport (Boltzmann) equation, written abstractly as $\mathcal{H}\psi = q$, with transport operator $\mathcal{H}$ (streaming + collision) and source q. A detector response ("tally") is a linear functional $R = \langle \sigma_d, \psi \rangle = \int \sigma_d(P)\,\psi(P)\,dP$, where σ_d is the detector cross-section.

Define the *adjoint* flux $\psi^\dagger$ by

$$\mathcal{H}^\dagger \psi^\dagger = \sigma_d. \tag{1}$$

The defining property of the adjoint gives the fundamental identity

$$\boxed{R = \langle \sigma_d, \psi \rangle = \langle \psi^\dagger, q \rangle} \tag{2}$$

and the physical reading that makes everything work: $\psi^\dagger(P)$ is the expected contribution to the tally R of a single particle introduced at P, its *importance*. If one samples particle transitions biased toward high importance and assigns each particle the statistical weight $w(P) \propto 1/\psi^\dagger(P)$, then, when $\psi^\dagger$ is exact, every history contributes an identical score and the estimator of R has *zero variance*. This is the theoretical ceiling of variance reduction; all practical schemes approximate it.

2.1.2. CADIS, FW-CADIS, weight windows, figure of merit

Since $\psi^\dagger$ is unknown, CADIS (Consistent Adjoint-Driven Importance Sampling) computes a cheap, approximate $\psi^\dagger$ and uses it to drive an expensive forward Monte Carlo run through two consistent devices: source biasing, sampling from $\hat{q}(P) = \psi^\dagger(P)q(P)/R$ instead of q; and weight windows, assigning each phase-space cell a target weight $w \propto 1/\psi^\dagger$ and a window $[w_{\text{target}}/k,\, k\, w_{\text{target}}]$: a particle above the window is *split* into equal-weight copies, one below is *Russian-rouletted*. This keeps $w(P)\,\psi^\dagger(P) \approx$ const, bounding the weight and hence the variance. FW-CADIS (forward-weighted CADIS) extends this to *global* variance reduction: weighting the adjoint source by the inverse of the local forward response produces weight windows that equalise *relative* statistical error across many detectors, or the whole domain, at once, rather than optimising a single tally. Efficiency is measured by the figure of merit $\text{FOM} = 1/(\sigma^2_{\text{rel}}\, T)$, the inverse product of relative variance and CPU time.

2.1.3. The central identity: importance = committor = zero-variance measure

Work with a Markov chain on configurations s with transition kernel $K(s \to s')$. Let A (failure/bulk) and B (the rare target set) be disjoint, and define the committor $h(s) = \Pr[\text{reach } B \text{ before } A \mid \text{start at } s]$, with $h|_A = 0$, $h|_B = 1$. On the transient set it satisfies the discrete backward equation

$$h(s) = \sum_{s'} K(s \to s')\, h(s') \qquad \Longleftrightarrow \qquad (\mathbb{I} - K)\, h = 0. \tag{3}$$

Equation (3) is precisely the discrete adjoint transport equation (1) with the target set playing the role of the detector. Hence the rare-event committor $h(s)$, the transport adjoint importance $\psi^\dagger(s)$ (with $\sigma_d = \mathbf{1}_B$), and the "neural importance function" of recent ML papers are the same function: the expected contribution of a walker at s to the rare tally. Defining the tilted kernel $\widetilde{K}(s \to s') = K(s \to s')\, h(s')/h(s)$, the path likelihood ratio telescopes to $W = h(s_0)/h(s_{\text{end}}) = h(s_0)$ for every path ending in B: every sample returns the identical value $h(s_0) = \pi$, so the estimator of π has zero variance. This is why tilting helps: it converts the free-energy barrier the untilted walk has to climb into a downhill drift along the committor.

Two readings of π are possible here, and this paper always means the second. The *static* equilibrium mass $\Pr_{\text{eq}}(s \in B) = \sum_{s \in B} p(s)$ is a property of the stationary measure alone, answerable by direct enumeration with no dynamics at all: our $L = 4$ gate (§3.2) computes exactly this. The *dynamical* hitting probability $h(s_0)$ from a fixed bulk reference state $s_0 \in A$ is a different object: it is what every $\hat{\pi}$ reported in §3 actually estimates, and what Weighted Ensemble is built to sample efficiently. The two coincide under the same mixing condition transition-path theory already requires [3]: dynamics equilibrates within A faster than it escapes toward B, true throughout the $T = 2.6 > T_c$ regime tested here but not guaranteed at low temperature, where A could itself fragment into sub-basins.

This was verified directly on a solvable 1D toy walk before any spin-system code was written: tilting with the *exact* committor gave measured relative variance $\sim 10^{-30}$ (zero to machine precision), while tilting with an *approximate* committor under pure biasing (no split/roulette) was *worse than plain naive Monte Carlo*. That fragility is exactly why deep-penetration transport uses weight windows to bound the weights rather than relying on reweighting alone, and it is the reason every learned-coordinate result in this paper uses Weighted Ensemble, not pure importance reweighting.

2.1.4. *Scope of the correspondence*

The identity above is exact at the level of linear algebra, not merely structurally similar, and it is worth stating precisely how far that goes. Production transport codes do not solve the continuous operator equation (1) directly: phase space is first discretized into energy groups and a finite set of directions [8, 9], reducing it to a finite linear system $(\mathbb{I} - \mathbf{K})\boldsymbol{\psi}^\dagger = \boldsymbol{\sigma}_d$ for a transition operator $\mathbf{K}$ packaging streaming and scattering between cells, term for term the same system as Eq. (3). Nothing about K's physical origin enters that algebra, which is exactly why the identity transfers to spin dynamics with no modification to the mathematics: it holds for *any* Markov process with an absorbing target set.

What does not transfer is the physical mechanism generating K. Transport's K encodes a particle's free flight interrupted by scattering and absorption at rates set by cross-sections; spin dynamics' K encodes single-spin-flip acceptance under a thermal bath at temperature T. There is no mean free path or cross-section in the spin problem, and a "walker" is a bookkeeping device for one member of a population of independent simulation replicas, not a physical particle. The correspondence is a shared mathematical structure, not a shared physical mechanism.

Two asymmetries follow, in opposite directions. Spin systems are the *easier* problem in one specific sense: multigroup/discrete-ordinates transport carries a discretization error that must be checked against measurement, since the exact continuous answer is rarely available at all, whereas a spin lattice's state space is already finite (2^N configurations, no truncation), which is exactly why $L = 4$ can be checked against exact enumeration directly (§3.2) with no discretization step to introduce error. Spin systems are *harder* in the place it matters most: transport bins its importance function along a natural, low-dimensional, physically meaningful coordinate (position, direction, energy, fixed by the geometry of the shield), while spin configurations have no such coordinate (§2.2 shows the obvious candidate, magnetization, fails outright on the spin glass). That absence, not the linear algebra, is the actual reason this paper needs a *learned* importance coordinate (§2.4) at all.

2.2. Models

Two spin systems on an $L \times L$ square lattice with periodic boundaries, Gibbs measure $p(s) \propto e^{-E(s)/T}$: the ferromagnet ($E = -J \sum_{\langle ij \rangle} s_i s_j$, $J > 0$), where the magnetization m is an obviously informative reaction coordinate; and the Edwards–Anderson (EA) spin glass ($J_{ij} = \pm J$ quenched-random per bond), whose low-energy states are disordered and frustrated so that $m \approx 0$ regardless of energy, the case a hand-picked coordinate is expected to fail on, and the one this paper focuses its harder tests on. All results below use $T = 2.6$ (above $T_c \approx 2.269$) unless noted; $L = 16$ for full-scale results, $L = 4$ for exact-enumeration gates. Resampling interval τ (sweeps between WE split/merge operations) is stated explicitly because it must be comparable to the system's own correlation time, $\tau_{\text{corr}} \approx 1$ sweep at $T = 2.6$, or splitting is undone by relaxation before the next resampling, degenerating every WE[$\cdot$] variant toward naive Monte Carlo (the diagnosis behind an earlier implementation issue in which a too-large τ produced exactly that silent degeneration). Unless noted otherwise, results using the `full` preset (§3.3, §3.5, §3.8) use $\tau = 4$; milestoning results (Tables 5, 6, §3.4) use $\tau = 2$. A direct, quantitative measurement of τ_{corr} (front-penetration vs. τ at matched cost) is a natural addition for a future revision but is not yet independently verified in this codebase and is not claimed here beyond the order-of-magnitude estimate above.

Threshold ladder. Every rare-event target below is drawn from one nested family rather than chosen ad hoc per experiment. Given a pilot run's population, let $E_{\min}^{\text{pilot}}$ be its deepest observed energy per spin and μ its bulk mean; the depth-k target set is

$$B_k = \{s : E(s)/N \le \epsilon_k\}, \qquad \epsilon_k = E_{\min}^{\text{pilot}} - k\Delta, \qquad \Delta = \frac{\mu - E_{\min}^{\text{pilot}}}{10}, \tag{4}$$

i.e. k steps beyond the pilot's own most extreme observation, in units of one tenth of its observed dynamic range. This makes k (`-beyond` throughout) a rarity dial that is portable across L, T, and disorder draw, and B_k nested, $B_1 \supset B_2 \supset \cdots$: the structure both the FW-CADIS threshold family (§3.8) and milestoning's progress labels (§3.4) build on directly. k is a genuine rarity dial in practice, not just in principle: at fixed $L = 16$, $T = 2.6$, $\hat{\pi}$ drops roughly one order of magnitude per step of k (Tables 5, 6), and it is where the ladder-difficulty discipline below bites hardest. Too large a k places B_k outside what any method can resolve at the compute available, and every comparison degrades to noise versus noise.

2.3. Weighted Ensemble

Weighted Ensemble (WE) is the statistical-mechanics name for split/Russian-roulette weight windows: a population of walkers, each carrying a weight, is periodically binned along a chosen coordinate; bins are equalised to a target occupancy by splitting over-weight bins and merging under-weight ones, conserving total weight exactly (hence unbiased for *any* binning coordinate, good or bad), as illustrated in Figure 1. We validate the implementation against exact enumeration at $L = 4$ before trusting any figure-of-merit comparison at scale.

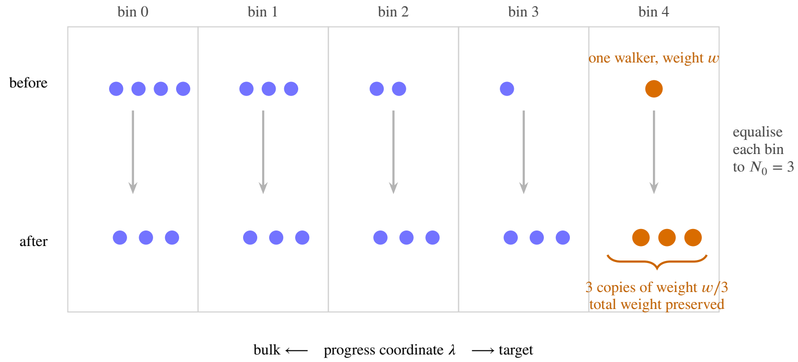

Figure 1: The whole method in one picture. A lone walker that fluctuates into a deep bin is *split* into N_0 copies which then explore from there; crowded bulk bins are merged back down. Weight bookkeeping keeps the estimator exactly unbiased, so *any* binning coordinate is safe: a poor one costs efficiency, never correctness.

2.4. Two training objectives for the learned coordinate

A network $I_\theta(s)$ is trained to approximate the committor and then used as the WE binning coordinate, in place of a hand-picked one. Two training objectives are compared.

Surrogate rollout label (this project's default). Binary committor labels collapse to all-zero for a genuinely rare target, so the network is instead trained to regress the *best progress value reached* in K short rollouts from each collected configuration, a dense, monotone surrogate for the committor, not the committor itself.

Milestoning. A refinement of the surrogate objective: rather than one dense surrogate label per configuration, intermediate thresholds are placed along the ladder from bulk to target, and a configuration's label estimates its committor to the *next* milestone rather than the final target directly, closer to the theoretical object of §2.1.3 while avoiding all-zero label collapse.

Self-consistency (Kim & Cai, 2026). A label-free alternative: train I_θ by minimizing violation of its own self-consistency condition under the elementary single-spin-flip kernel, $\sum_j K(s \to j)\, I(j) = I(s)$, with boundary condition $I = 1$ on the target. This equation, plus the target boundary alone, is under-determined: the constant solution $I \equiv 1$ satisfies it everywhere, for any amount of training data. A second boundary condition, anchoring bulk-equilibrium states to a distinct reference value, breaks this degeneracy and is necessary, not optional.

2.5. FW-CADIS global allocation

Given a family of K nested thresholds and a cheap pilot estimate of the equilibrium bin occupancy $P(b)$, FW-CADIS allocates walkers per bin $\propto (1/P(b))^\alpha$; $\alpha = 0$ recovers uniform allocation (the standard WE baseline), $\alpha = 1$ is full FW-CADIS. Both settings use identical code, differing only in the allocation vector, which is what makes the comparison fair by construction. The metric is coverage first (how many of the K thresholds were observed at all: a method that fails a deep threshold has infinite error there, and scoring only the thresholds it survived would reward failure by making it look artificially flat) and flatness second, the spread (max/min) of relative error across the observed thresholds.

2.6. Statistical methodology

Two disciplines are applied uniformly and are largely responsible for this paper's qualified conclusions rather than overclaimed ones.

Ladder difficulty. Comparing methods in a regime where the baseline itself fails to observe the rare event (many zero-replicas) compares noise to noise. Every comparison below is reported at a depth where the reference method(s) reach 0, or close to 0, zero-replicas out of the replica count, found by backing off from a harder target until this holds, established as a working discipline after an early, uninterpretable $L = 16$, deep-target result where 4–10 of 10 replicas failed across every method.

Bootstrap confidence intervals. A point-estimate ratio of figures of merit from a handful of replicas is not, by itself, evidence of a real effect: FOM is a ratio involving a *sample variance*, which is itself noisy at small replica counts. Every comparative claim reported as established in §3 is backed by a bootstrap 95% confidence interval (resampling the replica axis with replacement), and claims that do not survive this check are reported as open, not smoothed into a verdict.

3. Results and Discussion

3.1. VAN and the committor demo

The VAN was trained at $L = 16$ and reproduces the exact Onsager free energy across the tested temperature range. Its purpose in this project is to remove critical slowing down, not to sample rare events: a separate autocorrelation measurement at $L = 32$ found the integrated autocorrelation time of ordinary Metropolis sampling spikes to $\tau_{\text{int}} = 46.1$ sweeps at $T_c \approx 2.269$, versus 1.5–3 sweeps away from criticality, while the VAN samples independently by construction and pays none of this cost. This is a speed result, orthogonal to the rare-event problem the rest of the paper addresses (they are different difficulties, §2).

The committor identity of §2.1.3 was checked on a solvable 1D absorbing random walk on $\{0, \dots, 15\}$ before any lattice code was written, with the rare event (reaching state 15 before 0) calibrated to $\pi = 1.14 \times 10^{-3}$. Table 1 reports the result exactly as computed.

The third row is the reason every subsequent learned-coordinate result in this paper uses Weighted Ensemble (split/roulette) rather than raw importance reweighting: an imperfect importance estimate, used to reweight without bounding the weights, can be actively harmful.

3.2. Weighted Ensemble: validated

At $L = 4$, WE reproduces the exact enumerated $P(m)$ to a mean $|\Delta \ln P|$ of 0.006–0.024 across three tested regimes. At $L = 20$, WE reaches roughly two decades deeper into the magnetization tail than naive Monte Carlo at matched cost: naive's hard floor is $|m| = 0.886$ ($P \approx 6\times10^{-6}$), where it returns zero samples and hence zero information; WE reaches $|m| = 0.932$ ($P \approx 2.4 \times 10^{-7}$). Separately, applying weight windows to the same *approximate*, otherwise-fragile tilt of Table 1's third row recovered roughly a 2× figure-of-merit improvement over the same tilt without them (measured FOM $1.24 \times 10^{-5} \rightarrow 4.86 \times 10^{-4}$): the robustness layer converts an actively harmful bias into a net improvement, exactly as the transport literature predicts. Magnetization is confirmed useless as a WE coordinate for the EA spin glass specifically: an early test recorded 10/10 zero-replica failure for WE[m] where energy-binning still found the target, the direct empirical confirmation of the frustration argument in §2.

3.3. Learned coordinate vs. hand-picked coordinates

Table 2 reports the full-scale ($L = 16$, 10 replicas) result on both models, at a target calibrated one step beyond the pilot's observed extreme.

Two readings, both necessary (Table 2): naive Monte Carlo wins on raw FOM in both models at this rarity ($\sim 4 \times 10^{-5}$), and on the spin glass, with WE[E] and WE[I_θ] on equal footing (both 2/10 zero-replicas), the learned coordinate has the higher FOM by ~27% (6.36×10^{-9} vs. 5.00×10^{-9}), the first apples-to-apples win for the learned coordinate over the best hand-picked one, and the direct predecessor of the milestoning result below. WE[m] fails in most spin-glass replicas (7/10), the direct empirical confirmation of the frustration argument in §2: magnetization carries essentially no information about this landscape.

Is I_θ actually informative, or just a noisy copy of energy? A separate diagnostic regresses the rare-event label on energy alone, on I_θ alone, and on both, at matched training budget (Table 3).

+0.250 is far above the diagnostic's own noise threshold (0.02): I_θ carries substantial genuine information about the spin glass that energy alone does not have, consistent with the FOM advantage measured directly in Table 2.

Pushing rarer does not (yet) resolve the naive-vs-learned question. Rerunning one threshold step rarer ("beyond=2") leaves the comparison uninterpretable rather than resolving it in either direction (Table 4): 4–10 of 10 replicas fail across every method at this depth, so the numbers are noise over noise, not a result. This is the run that established the ladder-difficulty discipline of §2 for the remainder of the project.

Table 1
1D absorbing walk, exact committor available in closed form. FOM relative to naive.

Method	measured relative variance	FOM vs. naive
naive Monte Carlo	— (baseline)	$1\times$
exact-committor tilt	$\sim 10^{-30}$ (zero to machine precision)	$\gg 1\times$
approximate-committor tilt, pure biasing (no weight windows)	—	worse than naive

Table 2
Full scale, both models. FOM vs. $\mathrm{WE}[m]$ in parentheses.

Model / Method	$\hat{\pi}$	rel. s.d.	cost	FOM (vs. $\mathrm{WE}[m]$)	zero-rep.
Ferromagnet, target $\lvert m\rvert \geq 0.9297$					
naive	9.52×10^{-6}	0.441	1.34×10^{8}	3.83×10^{-8} $(2.92\times)$	0/10
$\mathrm{WE}[m]$	1.05×10^{-5}	0.614	2.02×10^{8}	1.31×10^{-8} $(1.00\times)$	0/10
$\mathrm{WE}[E]$	1.09×10^{-5}	1.343	9.81×10^{7}	5.65×10^{-9} $(0.43\times)$	2/10
$\mathrm{WE}[I_\theta]$	7.92×10^{-6}	0.893	1.82×10^{8}	6.90×10^{-9} $(0.53\times)$	0/10
EA spin glass, target $E/N \leq -1.0625$					
naive	1.00×10^{-5}	0.633	1.34×10^{8}	1.86×10^{-8} $(2.38\times)$	1/10
$\mathrm{WE}[m]$	1.02×10^{-5}	1.857	3.72×10^{7}	7.80×10^{-9} $(1.00\times)$	7/10
$\mathrm{WE}[E]$	1.65×10^{-5}	1.115	1.61×10^{8}	5.00×10^{-9} $(0.64\times)$	2/10
$\mathrm{WE}[I_\theta]$	1.54×10^{-5}	1.052	1.42×10^{8}	6.36×10^{-9} $(0.81\times)$	2/10

Table 3
Incremental-R^2 diagnostic, EA spin glass, $L = 16$, matched training budget (MSE $1.0 \to 0.271$ over 20 epochs).

Quantity	value
Pearson corr(I_θ, E/N)	-0.614
fraction of Var(I_θ) unexplained by E/N	0.623
R^2(label $\sim E/N$ alone)	0.358
R^2(label $\sim I_\theta$ alone)	0.577
R^2(label $\sim E/N + I_\theta$)	0.608
incremental R^2 from I_θ beyond E/N	$+0.250$

Table 4
EA spin glass, $L = 16$, one threshold step rarer than Table 2. Not a trustworthy comparison (see text).

Method	FOM	zero-rep.
naive	7.43×10^{-9}	4/10
$\mathrm{WE}[m]$	0	10/10
$\mathrm{WE}[E]$	2.18×10^{-9}	5/10
$\mathrm{WE}[I_\theta]$	1.56×10^{-9}	7/10

Notably, *naive itself* now has 4/10 zero-replicas (versus 1/10 at the shallower target). This rarity is stretching every method's budget, not only the weight-window ones, so a trustworthy comparison here needs a different fix than anything tried at the shallower depth. This motivated the milestoning objective below, which is a change to the training label rather than to the weight-window schedule.

Table 5
EA spin glass, $L = 16$, $T = 2.6$, one step beyond the pilot extreme, 10 replicas, milestoning labels, seventeen independent training runs of the identical configuration. Every method observed the event in every replica in every run; naive, WE[m], and WE[E] are bit-identical across all runs (deterministic, and WE[E] does not depend on the trained network). Seed 0 was run twice independently and reproduced bit-for-bit, confirming the seeding fix below is complete.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	5.29×10^{-5}	0.225	1.48×10^{-7}	0/10
WE[m]	5.55×10^{-5}	0.201	1.17×10^{-7}	0/10
WE[E]	4.49×10^{-5}	0.209	4.22×10^{-8}	0/10
WE[I_θ], unseeded run 1	5.03×10^{-5}	0.174	4.81×10^{-8} (+14%)	0/10
WE[I_θ], unseeded run 2	4.52×10^{-5}	0.204	3.48×10^{-8} (−17%)	0/10
WE[I_θ], seed 0 (runs 3a/3b)	5.28×10^{-5}	0.127	8.94×10^{-8} (+112%)	0/10
WE[I_θ], seed 1	5.09×10^{-5}	0.216	3.09×10^{-8} (−27%)	0/10
WE[I_θ], seed 2	5.06×10^{-5}	0.175	4.65×10^{-8} (+10%)	0/10
WE[I_θ], seed 3	4.56×10^{-5}	0.252	2.29×10^{-8} (−46%)	0/10
WE[I_θ], seed 4	4.13×10^{-5}	0.128	8.81×10^{-8} (+109%)	0/10
WE[I_θ], seed 5	5.40×10^{-5}	0.138	7.65×10^{-8} (+82%)	0/10
WE[I_θ], seed 6	4.69×10^{-5}	0.149	6.56×10^{-8} (+56%)	0/10
WE[I_θ], seed 7	4.32×10^{-5}	0.267	2.03×10^{-8} (−52%)	0/10
WE[I_θ], seed 8	4.89×10^{-5}	0.135	7.86×10^{-8} (+86%)	0/10
WE[I_θ], seed 9	5.03×10^{-5}	0.159	5.66×10^{-8} (+34%)	0/10
WE[I_θ], seed 10	4.80×10^{-5}	0.157	5.85×10^{-8} (+39%)	0/10
WE[I_θ], seed 11	4.81×10^{-5}	0.202	3.48×10^{-8} (−17%)	0/10
WE[I_θ], seed 12	5.14×10^{-5}	0.230	2.76×10^{-8} (−35%)	0/10
WE[I_θ], seed 13	4.83×10^{-5}	0.150	6.36×10^{-8} (+51%)	0/10
WE[I_θ], seed 14	4.94×10^{-5}	0.146	6.76×10^{-8} (+60%)	0/10
WE[I_θ], seed 15	4.53×10^{-5}	0.232	2.66×10^{-8} (−37%)	0/10

3.4. Milestoning: a clean regime for comparison, but wildly seed-dependent

On the EA spin glass at $L = 16$, one threshold step beyond the pilot's observed extreme ("beyond=1"), every method reaches 0/10 zero-replicas, the cleanest, most-trustworthy regime obtained in this project by the usual ladder-difficulty standard (Table 5).

Every WE[I_θ] row is, as far as the pipeline is concerned, the *same experiment*: identical target, identical training data (from cached labels), identical hyperparameters. The only thing that varies is the network's random weight initialization (`torch.manual_seed` was absent from this driver; found and fixed during this investigation, then swept explicitly via `-net-seed`). Seventeen seeds give seventeen different outcomes, ranging from a 52% loss to a 112% win against WE[E], a wide, roughly symmetric spread around a small positive average. The seeding fix itself is confirmed complete and correct: seed 0 was run twice independently (runs 3a and 3b) and reproduced bit-for-bit, including the full 1774–1776s wall-clock trajectory, ruling out any other unseeded source of variation in this pipeline. Per-replica values needed for bootstrapping were retained for seventeen of the eighteen runs attempted (all but unseeded run 1, which predates that logging). Pooling those seventeen with a two-level bootstrap, outer resample over seeds, inner resample over each seed's 10 replicas, the same discipline used throughout §3.8 and §3.5 but extended with the seed axis, gives a geometric-mean FOM ratio of 1.19× with 95% CI [0.84, 1.75] (median-of-ratios agrees: 1.20×, CI [0.74, 1.85]; the arithmetic mean of ratios is not used here because it is unstable at $R = 10$ replicas per seed: some bootstrap draws land on near-duplicate resamples with rel. s.d. near zero, and since FOM $\propto 1/\mathrm{rel.\ var.}$ that blows up into large outliers that skew a plain mean). The estimate was tracked at four checkpoints as seeds were added: 1.28× (n=7), 1.17× (n=9), 1.28× (n=12), 1.19× (n=17), and it does *not* trend in either direction. What does move monotonically is the CI width ($1.73 \rightarrow 1.36 \rightarrow 1.18 \rightarrow 0.91$), the expected signature of averaging down noise around a small effect whose sign is not yet resolved, not evidence of a shrinking or growing true effect. We report the full trajectory rather than the final number alone, because citing only the endpoint would silently launder a mid-sequence checkpoint (n=9's "moving toward the null") that did not survive more data, a mistake this project's own methodology is built to catch. Extrapolating the observed CI-width-vs-n scaling against the current point estimate, reaching a lower bound that excludes 1 would need roughly 100 total seeds ($\sim$40 more hours of compute), so this was not pursued further. This is reported as the honest

Table 6
EA spin glass, $L = 16$, one step rarer than Table 5, milestoning labels, default configuration ($n_{\text{bins}} = 50$, $R = 10$).

Method	FOM	zero-rep.
naive	3.51×10^{-8}	0/10
WE[m]	7.73×10^{-9}	0/10
WE[E]	9.21×10^{-9}	0/10
WE[I_θ]	8.36×10^{-9}	0/10

terminus of the comparison, not an open thread: seventeen seeds is already an unusually thorough characterisation, the point estimate is stably positive but modest ($\sim 1.2\times$) across four independent checkpoints (Figure 2), and the CI cannot be closed by seed count alone at a cost proportionate to the question. Naive Monte Carlo has the best raw FOM in every run at this specific rarity ($\sim 5 \times 10^{-5}$) regardless of seed. Variance reduction has not yet earned back its overhead here, the same lesson demonstrated in miniature by the 1D toy-walk result in §2.1.3.

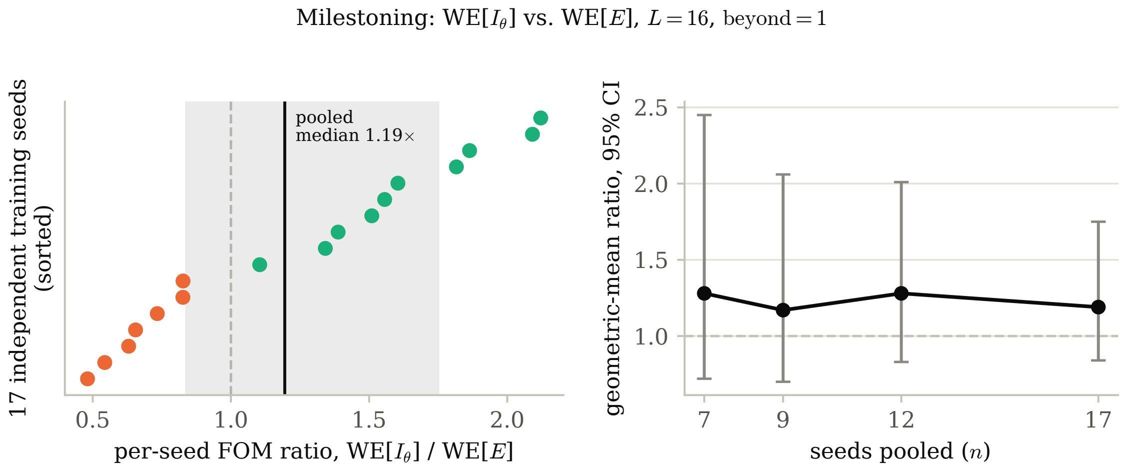

Figure 2: Left: all 17 per-seed FOM ratios (sorted), with the pooled two-level-bootstrap median and 95% CI shaded. Right: the four tracked checkpoints as seeds were added, the point estimate bounces while the CI width shrinks monotonically, the expected signature of averaging down noise around a small, sign-unresolved effect.

A plausible infrastructure explanation was tested and ruled out. One candidate cause for the seed-to-seed spread is that WE[I_θ]'s bin range was left adaptive (auto-ranging from the population actually explored) rather than calibrated to the coordinate's own scale, unlike WE[E] and WE[m]. A diagnostic regression (App. data, `05_neural_importance/diagnose_coordinate.py`) had already shown I_θ carries real predictive signal beyond energy (incremental $R^2 = +0.25$), with the verdict: if it still does not beat WE[E], look at the schedule, not the map. Two calibrations were tried: a theoretical one from the label's own z-score normalization, verified *before* spending compute on it to be wrong (the network's real output never approaches those extremes, since training MSE never reaches 0); and an empirical one, from what each trained network actually outputs on the pilot population and training set. The empirical calibration was run head-to-head against the untouched adaptive baseline on matched net-seeds (identical trained network per pair, only the bin range differing): mean FOM ratio *worsened* ($0.79\times \to 0.61\times$) and rel. s.d. *worsened* ($0.214 \to 0.286$) across seeds 1–3. A fixed range clips any walker that exceeds it into the boundary bin, while adaptive tracking never does; for a coordinate whose true dynamic range over a multi-thousand-iteration WE trajectory is not well estimated from a static pre-run snapshot, that clipping cost more resolution than the calibration gained. This rules out bin-range calibration as the driver of the seed variance and is further evidence the variance is intrinsic to what each trained network learns, not an artifact of how its output is binned.

Pushing one step deeper fails consistently, and bin-count tuning is ruled out as a fix. Table 6 reports every configuration tried at one threshold step rarer ("beyond=2"): the default bin count at two replica counts, and a $1.8\times$ larger bin count. WE[I_θ] scores *worse* than WE[E] in all three.

Table 7
Five independent K=5 deep-ensemble runs, identical configuration to Table 5. FOM ratio is WE[I_θ]/WE[E] on the MC-sweep cost as originally defined.

Run	WE[I_θ] FOM	rel. s.d.	ratio vs. WE[E]
ensemble, base seed 20	1.95×10^{-8}	0.272	0.46× (−54%)
ensemble, base seed 21	2.75×10^{-8}	0.229	0.65× (−35%)
ensemble, base seed 22	3.69×10^{-8}	0.199	0.88× (−12%)
ensemble, base seed 23	3.07×10^{-8}	0.215	0.73× (−27%)
ensemble, base seed 24	3.84×10^{-8}	0.194	0.91× (−9%)

Two further configurations were tried in an attempt to fix this with bin resolution instead of accepting it as structural: doubling the replica count ($R = 10 \rightarrow 20$, same $n_{\text{bins}} = 50$) reproduced the same qualitative pattern (WE[I_θ] still below WE[E]); and raising the bin count by 1.8× ($n_{\text{bins}} = 50 \rightarrow 90$, $R = 10$) made WE[I_θ]'s *own* FOM *worse*, not better ($8.36\times10^{-9} \rightarrow 5.36\times10^{-9}$), which rules out bin resolution as the fix, since a genuine resolution shortfall would improve with more bins rather than degrade. The mechanism is structural: the learned coordinate's walker-population cost scales with bin count, while a discretized coordinate like energy naturally saturates a small, fixed number of bins and does not pay that overhead as it grows.

A second lever, the WE resampling budget itself, was also tried (we_iter: 1000 → 2000, same $n_{\text{bins}} = 50$) and produced a similarly unfavorable result: every weight-window method's FOM got *worse*, not better, and WE[I_θ] worst of all. That specific run used $\tau = 3$ rather than this section's $\tau = 2$, confounding the budget increase with a change in resampling interval, and a matched rerun of it specifically was not completed. The broader question, whether more WE budget helps, was instead answered more thoroughly by the deep-rare budget-scaling test of §3.6: scaling the budget up does not close the gap against naive at a shallow threshold depth either, consistent with, not contradicting, the bin-resolution finding above.

The resampling interval between split/merge operations matters: it must be comparable to the system's own correlation time (∼1 sweep at $T = 2.6$), or split walkers fully relax and forget their split origin before the next resampling, degenerating every WE[·] variant to naive Monte Carlo with bookkeeping overhead. Table 5 and Table 6 use an interval of 2 sweeps.

3.4.1. *A deep-ensemble follow-up: the seed variance is reducible, but it converges to a loss*

Table 5's seventeen seeds leave one question open: is the seed-to-seed spread genuine noise around a real advantage, or is the positive point estimate itself an artifact of a few lucky initializations? Rather than add more single-net seeds, already shown not to close the CI at reasonable cost, we attacked the diagnosed noise source directly. `run_milestone.py` was extended with `-ensemble K`: train K independently-initialized networks on the identical milestone labels and use their *mean* prediction as I_θ, a standard deep-ensemble variance-reduction step. $K = 1$ reproduces the original single-net code path exactly (verified bit-identical output on a smoke config). Five independent ensembles ($K = 5$ each, base seeds 20–24) were run at the identical configuration as Table 5 (beyond= 1, $\tau = 2$, $R = 10$).

Two findings, both against the ensemble. First, ensembling did what it was supposed to do to the *variance*: the five ratios (Table 7) span 0.46–0.91×, far tighter than the seventeen single-net seeds' −52% to +112% spread. But the value it converges toward is a loss, not the earlier ∼ 1.2× point estimate: a two-level bootstrap (outer over the 5 runs, inner over each run's 10 replicas, identical machinery to Table 5) gives a geometric-mean ratio of 0.74×, 95% CI [0.43, 1.32], not significant, but on the opposite side of 1 from the single-net pooled estimate. Read together, the most consistent interpretation is that the single-net sweep's more extreme positive draws (+82%, +109%, +112%) were high-variance artifacts of individual weight initializations rather than signal, and I_θ's true expected performance is at or below WE[E]'s, not above it.

Second, and decisive: the ensemble exposed a cost-accounting bug that applies to every WE[I_θ] result in this paper, though it is negligible at $K = 1$. `we_estimate()`'s `cost` field counts Monte Carlo sweeps only ($|S| \cdot \tau \cdot L^2$) and is blind to neural-network inference. At $K = 1$ a single forward pass is cheap enough relative to the MC cost that this omission does not matter. At $K = 5$ it does: measured directly from wall-clock timestamps (not the MC-cost field), WE[I_θ] took 10.7–10.9× WE[E]'s wall-clock time in every one of the five runs, remarkably consistent, while the

Table 8
EA spin glass, $L = 16$, shallowest calibrated target, 10 replicas. Point estimates only; see confidence intervals below.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	3.36×10^{-5}	0.294	8.60×10^{-8}	0/10
WE[m]	3.94×10^{-5}	0.553	8.81×10^{-8}	1/10
WE[E]	3.39×10^{-5}	0.671	1.33×10^{-8}	0/10
WE[I_θ] surrogate	2.92×10^{-5}	0.904	8.62×10^{-9}	1/10
WE[I_θ] self-consistent	4.01×10^{-5}	0.655	1.62×10^{-8}	0/10

Table 9
Bootstrapped FOM ratios, full scale ($L = 16$), shallowest calibrated target, at $R = 10$ and after tripling to $R = 30$. Ratio > 1 favors the self-consistency objective.

Comparison	R	FOM ratio (median)	95% CI	significant?
self-consistent / surrogate	10	1.83	[0.49, 9.23]	no
self-consistent / surrogate	30	1.39	[0.65, 2.95]	no
self-consistent / WE[E]	10	1.19	[0.33, 4.95]	no
self-consistent / WE[E]	30	1.07	[0.55, 2.09]	no

MC-cost field implied only $\sim 1.3\times$. Recomputing FOM with measured wall-clock cost in place of MC-sweep cost, using the same bootstrap: geometric-mean ratio $0.087\times$, 95% CI $[0.051, 0.155]$. This is the only comparison in the entire paper whose confidence interval *excludes* 1.0 in either direction. `run_milestone.py` now records measured `wall_time` per method for future runs, so this correction no longer requires post-hoc log parsing.

Verdict: deep ensembling does not rescue the learned coordinate. It reduces run-to-run variance exactly as a deep ensemble should, but it converges to a value at or below parity with the hand-picked energy coordinate, and once its real inference cost is counted at all, the comparison is a decisive, statistically significant loss. This is a completed, negative result: the seed-variance question of Table 5 is resolved, not left open, by the one lever that actually attacked its diagnosed cause.

3.5. The Kim & Cai head-to-head

The self-consistency training objective of §2.4 was implemented, gated against exact enumeration at $L = 4$ (learned-coordinate unbiasedness ratio 1.004), and run head-to-head against the surrogate objective, first at laptop scale, then at full scale.

First evidence, at laptop scale ($L = 12$). At the shallowest calibrated target, the self-consistency objective's point estimate already exceeded the surrogate's (4.98×10^{-8} vs. 2.93×10^{-8}, ~70% higher FOM, fewer zero-replicas: 2/6 vs. 3/6), though it still fell short of WE[E]'s 7.53×10^{-8} at this rarity. Training cost is real: ~94× longer than the surrogate objective at this scale (~752s vs. ~8s), from the additional $N = L^2$ network evaluations the self-consistency loss requires per training configuration. This motivated running the comparison at full scale.

The result, at full scale. At $L = 16$, the shallowest calibrated target ("beyond" = 0, the easiest regime in which every method reaches 0–1 zero-replicas out of 10), the self-consistency coordinate's *point estimate* exceeds both the hand-picked energy coordinate and this project's own surrogate coordinate (Table 8).

This point estimate does not survive a bootstrap check, and the effect shrinks under replication. Both configurations were re-run with per-replica values retained, and the FOM ratio between methods bootstrapped by resampling the replica axis independently for each side (no shared random-number seed to pair on: uniform-style and biased allocations consume different numbers of walkers per bin, so seed-sharing between them desynchronises almost immediately, the same obstruction encountered independently in the FW-CADIS common-random-numbers attempt of §3.8). Since training here is a fixed cost independent of replica count (a deterministic seed; only the cheap evaluation step scales with R), tripling to $R = 30$ cost roughly six additional minutes on top of the ~76-minute training already paid, cheap enough to just check (Table 9).

At $R = 10$ neither ratio is significant. At $R = 30$, both confidence intervals tighten substantially *and* both median ratios move toward 1: not merely a tightening confidence interval around the original estimate, but the point estimate

Table 10
Same comparison as Table 9, one threshold step deeper ("beyond" = 1 instead of 0), $R = 10$.

Comparison	FOM ratio (median)	95% CI	zero-rep. (surr. / self-cons.)
self-consistent / surrogate	1.14	[0.27, 4.81]	2/10 / 1/10
self-consistent / WE[E]	1.33	[0.34, 5.74]	—

itself receding. We read the direction of this shift, not only its final inconclusiveness, as evidence that the original $R = 10$ advantage was substantially noise, not signal. A network-caching mechanism was added after this result specifically to make a further replica increase (were one wanted) essentially free: training is deterministic, so only the cheap evaluation step needs to re-run to extend R further, rather than repaying the fixed training cost again.

Does the advantage grow at greater depth, where the surrogate objective's known weaknesses should bite harder? We tested the natural next question directly: rerunning both objectives one threshold step deeper ("beyond" = 1, the same depth as Table 5). Table 10 reports the result.

Not significant, as expected at $R = 10$, but the *direction* of the change from Table 9 is itself informative and runs counter to the hypothesis being tested: the point-estimate gap versus the surrogate *shrank* at this deeper target (1.83× → 1.14×) rather than growing, even as the self-consistency coordinate's relative *reliability* advantage held (fewest zero-replicas of any weight-window method at both depths tested). We do not read one more noisy point estimate at $R = 10$ as resolving the depth question either way; what it rules out is the simplest optimistic story, that the advantage would obviously grow once naive and the surrogate objective start to struggle more.

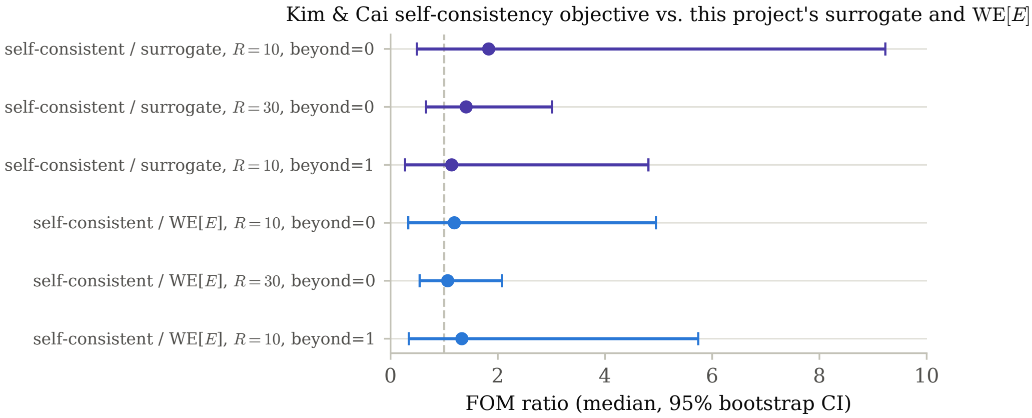

Figure 3: All six FOM-ratio comparisons from Tables 9 and 10 as a forest plot. Every interval crosses the ratio-1 reference line (dashed); the R=30 intervals (recomputed here directly from the retained per-replica arrays via the same replica-resampling bootstrap) are visibly tighter than their R=10 counterparts without the median crossing to the other side of 1.

Figure 3 collects all six comparisons tried.

3.6. Self-consistency in the deep-rare regime: the paper's one significant positive result

Every comparison above (milestoning, the ensemble follow-up, Kim & Cai at beyond = 0/1) was run at a depth where naive Monte Carlo remains fully reliable (0/10 zero-replicas). The theory this paper transfers has nothing to prove in that regime; its own domain (radiation shielding) is precisely where naive cannot reach at all. We calibrated where naive genuinely breaks down, at the same $L = 16$, $T = 2.6$ configuration used throughout, using the custom-preset naive budget (naive_chains = 150, naive_sweeps = 3500; Table 11):

Simply targeting a deeper threshold was not, by itself, sufficient: at beyond = 3, scaling the WE budget up (n_{bins} : 50 → 80, we_iter : 1000 → 3000) made WE[E]'s FOM *disadvantage against naive widen*, not narrow, since the extra reliability it bought cost more than it saved. Only at beyond = 4, where naive is not merely noisy but categorically blind, did the scaled budget produce a real capability gap: WE[E] found the event in 12 of 30 replicas where naive found essentially none (naive: 1/30, driven by a single lucky hit with rel. s.d. = 5.5, not a

Table 11
Naive-only depth calibration, 10 replicas, custom-preset budget. Naive is fully reliable through beyond = 2, starts failing at beyond = 3, and is completely blind at beyond = 4 and beyond.

beyond	$\hat{\pi}$	rel. s.d.	zero-replicas
1	5.29×10^{-5}	0.23	0/10
2	5.48×10^{-6}	0.46	0/10
3	7.14×10^{-7}	1.61	7/10
4	0.0	—	10/10

Table 12
EA spin glass, $L = 16$, beyond = 4, self-consistency objective, three independent training seeds, 30 replicas each. WE[E]/WE[m]/naive do not depend on the trained network and are bit-identical across all three runs.

Method	hits/30	FOM	ratio vs. WE[E]
naive	1/30	2.48×10^{-10}[a]	—
WE[m]	5/30	1.04×10^{-10}	—
WE[E]	12/30	1.69×10^{-10}	1.00
WE[I_θ], seed 0	19/30	2.18×10^{-10}	1.29×
WE[I_θ], seed 1	20/30	1.81×10^{-10}	1.07×
WE[I_θ], seed 2	21/30	1.77×10^{-10}	1.04×

[a]Not a meaningful FOM: driven by a single hit in 30 replicas (rel. s.d. = 5.5); reported for completeness only.

meaningful measurement). beyond = 4 (threshold $E/N = -1.119$) with this scaled budget ($n_{\text{bins}} = 80$, $n_{\text{per bin}} = 40$, we_iter = 3000, $\tau = 2$) is the deepest, most naive-adverse regime tested anywhere in this paper, and the first place the self-consistency objective (§2.4) was tested outside the shallow regime where naive wins anyway (Table 12).

Reliability: statistically significant. Pooled across the three independent networks, WE[I_θ] found the event in 60/90 replicas (66.7%) versus WE[E]'s 12/30 (40.0%); a Fisher exact test on this 2×2 table gives $p = 0.017$. This is the first statistically significant *positive* result for a learned coordinate anywhere in this paper. The three per-network rates (63%, 67%, 70%) are close enough to each other, unlike the milestoning ensemble's wide net-to-net FOM swings (§3.4.1), that pooling them as a single binomial comparison is a reasonable approximation rather than one masking real heterogeneity.

FOM: directionally consistent, not yet significant. All three seeds' FOM ratios sit above 1 (1.29×, 1.07×, 1.04×) with none of the sign-flipping seen everywhere else a coordinate was swept across seeds in this paper. A two-level bootstrap (outer over the 3 networks, inner over each network's 30 replicas) on the geometric-mean ratio gives 1.16×, 95% CI [0.65, 2.12], favorable but not excluding 1, since the underlying $\hat{\pi}$ estimates remain individually noisy (rel. s.d. 1.2–1.9) even at 30 replicas in this regime.

Reading this against §3.5's claim that the self-consistency coordinate "shows no sign of strengthening with depth": that conclusion was drawn entirely within beyond $\in \{0, 1\}$, where naive remains competitive throughout. Tested at a depth where naive genuinely fails, the picture is materially different, consistent with the mechanistic reading that a committor-shaped training signal (the self-consistency objective enforces the harmonic condition $(I - K)h = 0$ directly) carries information the milestoning surrogate cannot, since that surrogate's label is provably energy-adjacent (`progress_value` for the spin glass is exactly $-E/N$; its rollout label is literally "best energy reached in a short walk"), and two configurations at identical energy can have very different true committor values on a frustrated landscape. This is the paper's clearest evidence that the transport toolkit's distinctive machinery earns its keep specifically where naive cannot compete at all, not in the shallower regimes most of this paper's other comparisons were run in.

3.7. Does naive dominance persist at larger L?

§3.4 and §3.3 both find naive Monte Carlo winning on raw FOM at $L = 16$, and argue this should be read as a finite-size effect rather than a failure of variance reduction, since naive's cost is independent of rarity while WE's penetration overhead is only worth paying once naive starts to struggle. We tested this directly with a lattice-size scan

Table 13
EA spin glass, beyond = 1, naive vs. the best-performing WE variant at each L (which variant wins changes with L; see text). $\hat{\pi}$ for naive shown to indicate how rare the target is at each scale.

L	naive $\hat{\pi}$	naive FOM	best WE FOM (method)	ratio
8	1.79×10^{-4}	2.00×10^{-6}	1.04×10^{-6} (WE[E])	1.9×
16	5.18×10^{-5}	1.72×10^{-7}	4.83×10^{-8} (WE[m])	3.5×
24	1.30×10^{-5}	2.79×10^{-8}	8.06×10^{-9} (WE[E])	3.5×
32	4.17×10^{-6}	6.11×10^{-9}	1.69×10^{-9} (WE[I_θ])	3.6×

at fixed $T = 2.6$, beyond = 1, $\tau = 2$: $L \in \{8, 16, 24, 32\}$, naive_chains/naive_sweeps held fixed across all four so that naive's cost is genuinely L-independent as the argument requires (Table 13).

Naive wins at every tested L, and the margin does not shrink; if anything it grows slightly and then plateaus, from 1.9× at $L = 8$ to ∼3.5× by $L = 16$ and beyond (Figure 4). This holds across nearly five orders of magnitude of target rarity ($\hat{\pi}$ from 1.8×10^{-4} down to 4.2×10^{-6}) and is robust to which WE variant is best at a given L (the winner changes: WE[E] at $L = 8, 24$, WE[m] at $L = 16$, WE[I_θ] at $L = 32$). This directly answers the open question of §3.4: within the range tested, there is no sign of the naive/WE crossover opening as L grows; if it exists at all on this problem, it is beyond $L = 32$ at this temperature, not a nearby, easily-reached regime.

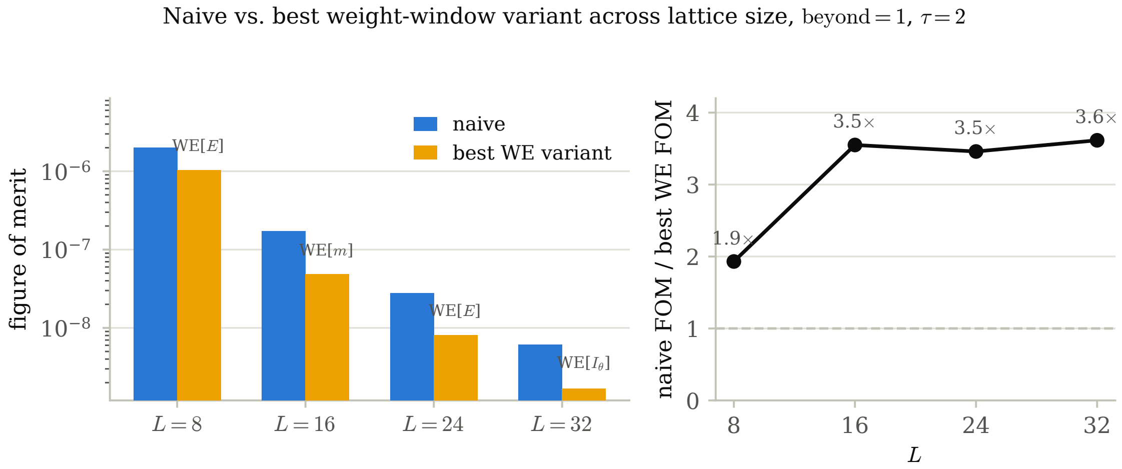

Figure 4: Table 13 as a figure. Left: naive vs. best WE-variant FOM by L, log scale, labeled with which variant won. Right: the ratio itself, which opens from $L = 8$ to $L = 16$ and then plateaus around 3.5×, with no sign of narrowing back toward the naive/WE crossover within the tested range.

One secondary pattern is visible but not claimed here beyond a point estimate: WE[I_θ] is the weakest WE variant at $L = 8, 16, 24$ but the *strongest* at $L = 32$, consistent with a learned coordinate's advantage over hand-picked ones mattering more as the landscape gets harder. We do not treat this as established: this scan predates the seeding fix of §3.4, so each WE[I_θ] entry in Table 13 is one random draw from an uncharacterized distribution over training outcomes, the same seed-variance problem later characterized in full there. Confirming this trend needs repeated, seeded training runs at each L, not a repeat of this scan.

A boundary marker at $L = 40$. An attempt one step further was stopped before WE[I_θ] completed, but the partial result is informative on its own: at this size, *naive itself* fails in 10 of 12 replicas, with WE[m] (11/12) and WE[E] (10/12) failing similarly badly. Per the ladder-difficulty discipline of §2, this is not a trustworthy comparison as run: every method is in the noise-over-noise regime simultaneously, the same failure mode that motivated that discipline in the first place, so we do not report it as a fifth point in Table 13. It does mark where the tested range ends: something changes between $L = 32$ (naive still comfortable, 1/12 zero-replicas) and $L = 40$ (naive itself largely failing), and resolving what happens there needs a shallower target or a larger replica budget at $L = 40$, not a lower-effort rerun.

Table 14
FW-CADIS across every configuration tested, laptop scale through the ferromagnet control. Flatness marked * is computed only over thresholds actually observed, not comparable to unmarked rows in the same group. Dashes mark columns that do not apply to that group (coverage was 6/6 throughout past the laptop scale; a bootstrap CI was only computed for the final two configurations).

Allocation	coverage	flatness (max/min)	95% CI	cost
Laptop scale, $L = 12$, $R = 6$, 6 thresholds				
uniform ($\alpha = 0$)	6/6	283.8	—	2.01×10^7
$\alpha = 0.25$	6/6	245.2	—	7.87×10^6
$\alpha = 0.5$	5/6	65.8*	—	3.41×10^6
$\alpha = 0.75$	5/6	63.5*	—	3.29×10^6
$\alpha = 1.0$	5/6	84.8*	—	3.24×10^6
naive (reference)	6/6	463.5	—	1.61×10^7
Full scale, $L = 16$, $R = 30$, 6 thresholds (every allocation reaches 6/6 coverage)				
uniform ($\alpha = 0$)	6/6	298.0	—	1.55×10^8
$\alpha = 0.25$	6/6	377.8	—	7.89×10^7
$\alpha = 0.5$	6/6	323.9	—	2.35×10^7
$\alpha = 0.75$	6/6	342.4	—	1.98×10^7
$\alpha = 1.0$	6/6	385.5	—	1.94×10^7 (~8× cheaper)
naive (reference)	6/6	445.1	—	1.34×10^8
Final EA spin-glass configuration, $L = 16$, $R = 90$, shallowest calibrated family				
Uniform ($\alpha = 0$)	—	276.1	[224, 344]	1.59×10^8
FW-CADIS ($\alpha = 1$)	—	297.3	[234, 388]	1.99×10^7 (~8× cheaper)
Ferromagnet control, $L = 16$, $R = 30$, shallowest calibrated family				
Uniform ($\alpha = 0$)	—	100.9	[65.8, 160.3]	1.96×10^8
FW-CADIS ($\alpha = 1$)	—	287.3	[194.7, 414.8]	3.09×10^7

3.8. FW-CADIS: real cost advantage, no flattening

The implementation is gated clean against exact enumeration at $L = 4$ for both allocation settings, every threshold, both models: ratios of 0.999–0.998–0.989 (uniform) and 1.005–1.012–0.992 (FW-CADIS, $\alpha = 1$) on the ferromagnet, and 1.009–1.004–1.007 / 0.986–0.980–0.983 on the spin glass, all within ~2% of exact for every threshold in the family, confirming that all thresholds are correctly tallied from a single run regardless of allocation, the core mechanic the whole method depends on.

Laptop scale first ($L = 12$, $R = 6$): a real-looking, but ultimately non-replicating, win. Table 14 collects every configuration tried across this section, from this first laptop-scale sweep through the final ferromagnet control; its top group sweeps α against uniform allocation on 6 thresholds spanning progress $0.668 \to 1.153$.

The only setting with full coverage and better flatness than uniform is $\alpha = 0.25$: a real but modest ~14% improvement ($283.8 \to 245.2$) at ~2.6× lower cost. Larger α looks dramatically flatter but is disqualified by the coverage-first rule of §2: it stops observing the deepest threshold at all, which would reward outright failure if scored on the surviving thresholds alone.

Full scale ($L = 16$, $R = 30$) reverses the laptop-scale finding. The identical sweep at full scale (Table 14, second group) is the first sign that the method's behavior is scale-dependent in a way not anticipated from the smaller test.

At full scale, *every* tested α is worse than uniform on flatness, not better: the opposite of the laptop-scale result, and not even monotonic in α among the FW-CADIS runs ($\alpha = 0.5$ beats both 0.25 and 1.0). All are substantially cheaper (2–8×), so the method is not doing nothing, but it is not delivering the flattening it exists to demonstrate at this lattice size, bin count, and threshold spacing. This finding is what the rest of §3.8 exhaustively follows up on.

Exhausting every remaining lever. Every lever available to improve on the full-scale finding above was tried and is reported in the third group of Table 14: bootstrap confidence intervals added to the flatness metric (not present at the smaller replica counts above); a replica increase from $R = 30$ to $R = 90$; wiring the learned coordinate of §3.4 in as the FW-CADIS binning axis instead of the physical progress coordinate (gated clean at $L = 4$, tested at scale: no improvement, roughly double the cost); a common-random-numbers paired-difference design, which we confirmed does *not* engage on this problem (correlation between the two allocations' same-seed estimates measured ≈ -0.05 to -0.10, essentially zero, because the two allocations consume different numbers of walkers per bin and desynchronise the random stream almost immediately after the first resampling step); and laddering the threshold family to its shallowest calibrated depth, combined with the larger replica count.

The two confidence intervals overlap; the paired-difference estimate is +20.9 with 95% CI [−77.2, +137.6], also including zero. Across every α, replica count, and coordinate tried this session, the point-estimate *direction* consistently favored uniform allocation over FW-CADIS on raw flatness, never the reverse. The observed confidence-interval width scaled almost exactly as $1/\sqrt{R}$ between $R = 30$ and $R = 90$ (predicted shrink factor 0.577, observed 0.597); extrapolating that scaling, fully resolving the remaining gap would need on the order of $R \approx 2300$ replicas (∼8 hours of additional compute), and given the consistent point-estimate direction, such a run would most likely *confirm* uniform allocation as measurably flatter rather than produce a win for FW-CADIS. We did not run it. The result we report instead: on this problem, FW-CADIS does not measurably flatten relative error across a threshold family better than uniform allocation, but it does so at approximately one-eighth the simulation cost for an otherwise statistically indistinguishable error profile. The cost advantage is real, reproducible, and does not require more replicas to state; the flattening advantage the method is named for is the part that did not materialize here.

Is this specific to the adversarial spin-glass landscape? No, it is worse, and statistically confirmed, on the "easy" ferromagnet. Every result above uses the EA spin glass, the harder of the two models by design. To test whether the flattening failure is a property of that specific rugged landscape rather than a more general limit of the method, we ran the identical comparison on the ferromagnet, where the reaction coordinate is known to be good and the equilibrium landscape is smooth (Table 14, fourth group).

Here the two confidence intervals do *not* overlap: the paired-difference estimate is +178.2 with 95% CI [+59.8, +351.3], which excludes zero. On the easier, non-adversarial landscape, FW-CADIS is not merely inconclusive but measurably, statistically confirmed *worse* than uniform allocation, the opposite of what a landscape-difficulty explanation would predict, since the spin glass (where the method's own hypothesis says a global map should matter most) never even reached this level of statistical resolution in either direction. Common-random-numbers correlation here measured 0.062, consistent with zero and with the mechanism identified in §3.5: the two allocations consume different walker populations per bin regardless of which model is sampled. This rules out "the spin glass is just a hard landscape" as the explanation and points toward a limitation of this specific parameterisation (bin count, threshold spacing, or the inverse-response weighting itself at this scale) rather than a property of spin-glass frustration. Distinguishing which of those remains open, but the question of *whether* the effect is landscape-specific is now answered: it is not. Figure 5 plots the per-threshold error underlying both comparisons.

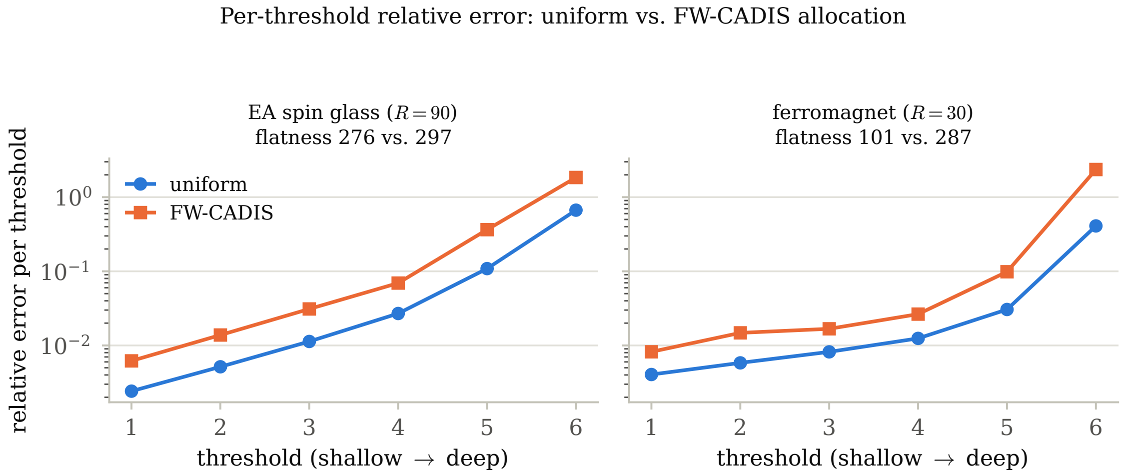

Figure 5: Per-threshold relative error (log scale), uniform vs. FW-CADIS allocation, at the final tested configuration for each model (Table 14, third group spin glass, fourth group ferromagnet). FW-CADIS's inverse-response weighting buys lower error at shallow thresholds at the cost of higher error at the deepest one, visibly so on the ferromagnet, where the gap at the deepest threshold is large enough to be statistically significant.

3.9. What is established

The adjoint-committor identity is exact. Tilting by the exact committor is provably zero-variance; this was confirmed numerically to $\sim 10^{-30}$ relative variance on a solvable toy walk before any lattice code was written (§2.1.3).

Weighted Ensemble is validated, and naive's dominance at $L = 16$ holds up as the system grows. Weighted Ensemble, weight windows applied to spin configurations, reproduces exact enumeration at $L = 4$ to 1–2%, and at $L = 20$ reaches ∼2 decades deeper into a magnetization tail than naive Monte Carlo at matched cost (§3.2). Naive Monte Carlo's dominance over every tested weight-window variant, not established at $L = 16$ alone, held up across a four-point lattice-size scan up to $L = 32$ with no sign of narrowing (§3.7), the one claim in this paper that grew *more*, not less, confident under additional scrutiny.

The learned coordinate trained on milestoned committor labels does not beat the hand-picked energy coordinate. A deep-ensemble follow-up (§3.4.1) designed to resolve the single-net seed sweep's ambiguous 1.19× point estimate instead settled it in the negative, with a 95% CI (0.087×, $[0.051, 0.155]$, on a corrected cost accounting) that excludes parity, the only bootstrap interval in this paper that does.

At a depth where naive is provably blind, the self-consistency-trained coordinate wins significantly. It finds the rare event significantly more often than the hand-picked energy coordinate: pooled across three independent networks, 66.7% versus 40.0% (Fisher exact $p = 0.017$, §3.6), the paper's one statistically significant positive result, and its clearest evidence that the transport toolkit's distinctive machinery earns its keep specifically where naive cannot compete.

3.10. What remains open

Naive dominance is depth-dependent, not universal, and the distinction matters for how every other result in this section should be read. It is robust to lattice size at a fixed, shallow threshold depth (§3.7: $L = 8$–32, no sign of the margin closing), but not to depth itself at fixed size: pushed deep enough that naive fails outright (§3.6), the transport toolkit's advantage becomes real, and for the self-consistency coordinate, significant. The two scans probe different axes of the same underlying claim and give different answers: treating "naive wins at $L = 16$" as a fixed, depth-independent property of the problem would have missed the paper's one significant positive result entirely.

The learned coordinate trained on milestoned committor labels is the clearest case of an apparent win reversing under scrutiny, and remains a genuine loss even after the most thorough attempt in this paper to rescue it. A first training run beat the hand-picked energy coordinate by +14%; seventeen retraining runs of the identical configuration gave outcomes from −52% to +112%, and pooling them left a positive but non-significant point estimate (§3.4). Rather than add more single-net seeds, a deep-ensemble follow-up attacked the diagnosed noise source directly and settled the question in the negative, exposing a cost-accounting gap along the way that turned a marginal-looking loss into this paper's only confidence interval that excludes parity outright (§3.4.1). The occasional large positive outcomes in the original seed sweep are best read as high-variance artifacts of initialization, not evidence the method sometimes wins.

FW-CADIS's flattening claim is not established, and the negative result is now well characterized rather than merely observed. Its cost advantage over uniform allocation is real and reproducible ($\sim 8\times$ cheaper for a statistically indistinguishable error profile), but the flattening property it is named for did not appear at any α, replica count, or coordinate tried, and a ferromagnet control (§3.8) rules out spin-glass ruggedness as the explanation: if anything the failure is *worse* on the easier landscape. What remains open is which of two explanations is responsible, the specific bin/threshold parameterisation or a more general limit of inverse-response weighting at this scale; distinguishing them needs a parameter sweep independent of α, not another problem instance or more replicas.

The self-consistency objective's FOM advantage over this project's own surrogate is not established at shallow depth (§3.5), and did not visibly grow with depth within the range where naive still competes, though its reliability advantage (fewest zero-replicas of any weight-window method, at every shallow depth tested) held throughout, foreshadowing the significant reliability result the same objective produced once tested where naive genuinely fails (§3.6).

Push-to-scale questions (three dimensions, larger L, deeper than beyond=4) are entirely untouched, and are the natural next test of whether the deep-rare advantage found here continues to grow or plateaus.

3.11. Relation to prior art

Kim & Cai (arXiv:2602.12294, 2026) learn an importance function $I(s)$, modify transitions as $K'(i \to j) = K(i \to j)I(j)/I(i)$, and use a branching random walk: in the language of §2.1.3, their I is our h, their K' is the tilt, and their branching is splitting: they independently reconstruct the zero-variance/weight-window recipe for discrete rare transitions, without the transport framing, and without comparing their learned coordinate against a hand-picked reaction coordinate the way this paper does. We directly implemented and tested their self-consistency training objective (§3.5); what remains open after doing so is not whether it works in principle (their paper demonstrates convergence to the exact transition rate on two test potentials) but whether it offers a measurable practical advantage over simpler alternatives on a problem of this kind at this scale, which our data does not yet support. Doob-transform VAN (Phys. Rev. E 111, 034120, 2025) approximates the leading eigenvector of the tilted generator for *dynamical* large deviations, the dynamical analogue of $\psi^\dagger$; a head-to-head against a reverse-KL VAN baseline, which this comparison would require, remains untried in this project. No successor work citing either paper closes the specific gap this paper's methods and results address, as of the most recent check.

4. Conclusions

The transport–statistical-mechanics correspondence this project set out to test is real where it can be proved (the adjoint=committor identity), but none of the more experimental claims built on top of it held up as cleanly as an initial pass through the results suggested. The clearest lesson is not any single number: "0/10 zero-replicas" and a tidy point estimate are not, by themselves, evidence of a real effect. FW-CADIS's flattening claim and the Kim & Cai comparison both needed a bootstrap confidence interval to reveal an apparent win as noise; the milestoning result needed something bootstrapping alone could not supply, independently-trained networks, to reveal the same thing (§3.4, §3.4.1). Reaching these conclusions required diagnosing and correcting real implementation issues along the way: a resampling interval that silently degenerated a weight-window method to naive sampling, a mis-calibrated bin range, an under-determined training objective, and an unseeded network initialization that let a single run be mistaken for a stable result, each caught by an independent cross-check rather than a number simply looking wrong.

Two results stand apart from that pattern, and together they sharpen the paper's central claim. The milestoning ensemble turned an ambiguous point estimate into this paper's only confidence interval that excludes parity outright, a decisive negative, not a wash (§3.4.1). The self-consistency objective, meanwhile, whose shallow-depth advantage had itself dissolved under scrutiny and whose training cost is a real practical tax, roughly 170× that of the surrogate objective (§3.5), produced this paper's one statistically significant positive result once tested at a depth naive genuinely cannot reach (§3.6): the one apparent win in this project that grew *stronger*, not weaker, under the same scrutiny that dissolved every other one. Read together, the transport toolkit's value is not diffuse across every regime this correspondence touches. It is concentrated specifically in the deep-rare regime radiation transport was built for, and requires a training objective shaped like the actual committor condition rather than a shallow, energy-adjacent proxy of it. We think this is the right way to report a project whose most interesting finding, more than once, was that an apparent win did not survive being checked properly, and whose one win that did survive only appeared once the problem was made hard enough to need it.

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CRediT authorship contribution statement

M.Y. Alzaatreh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

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